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PALAEOMAGNETISM OF MIDDLE ORDOVICIAN VOLCANIC ROCKS FROM QUEBEC

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This palaeomagnetic study is centered on agglomerates and volcanic rocks from the western margin of the Appalachian belt in the Drummondville–Actonvale–Granby area, Quebec (long.: 72°30'W, lat.: 46°00'N). It involves a total of 36 oriented samples (111 specimens) distributed over eleven sites. Both thermal and AF cleaning techniques were used to isolate residual remanent components. The dispersion of the directions is slightly reduced after AF cleaning and thermal treatment.

The palaeopole position obtained is 191°E, 6°N ($d_m = 14°, d_p = 7°$) after thermal treatment and 164°E, 19°N ($d_m = 11°, d_p = 6°$) after AF cleaning. The polarity of most of the sites (two exceptions) are reversed. The thermal-treated data appear to be relatively stable and an approximate value of the primary magnetization is extracted from them. The palaeopole obtained does not lie close to the tentatively defined position of the Cambrian and Ordovician poles from rocks of the North American plate; it is located near the Upper Cambrian and Lower Ordovician poles from eastern Newfoundland and the Lower Ordovician pole from the Caledonides in Europe.

1. Introduction

Palaeomagnetic data for Cambrian and Ordovician rocks from North America are not abundant (McElhinny, 1973). The Cambrian and Ordovician are represented by six and five palaeomagnetic pole positions respectively. The positions of these poles and their references are given in Table I. Five out of six Cambrian data and four out of five Ordovician data have been obtained from sedimentary rock sequences. The natural remanent magnetization (NRM) of sediments is generally quite complex in view of components of magnetization acquired after deposition (DRM) or later, by chemical alteration of the Fe oxides (CRM), or by heating of the magnetic oxides (PTRM). These secondary magnetizations explain the large scatter in the results from Cambrian and Ordovician age.

More palaeomagnetic studies of igneous rocks of Cambrian and Ordovician age are needed thus. In this study, palaeomagnetic data obtained from extrusive igneous rocks of the Appalachian front of southern Quebec are presented.

2. Geology

2.1. Stratigraphy and petrology

The rocks of the Drummondville–Actonvale–Granby region have an Early Palaeozoic age with the exception of a lamprophyre dyke related to the Monteregian Hills in the St.-Hyacinthe area, which is of Mesozoic age (whole-rock K–Ar dating by Shell Oil Co.: 116 ±8 m.y., i.e. Cretaceous). Fig. 1 shows an index map of the main localities and accompanying detailed geological maps of the Actonvale and Drummondville areas as well as the location of most of the sampling sites. The rocks are of sedimentary origin except for a few volcanic units. The first geological

TABLE I

Cambrian and Ordovician poles from North America

No.	Rock unit	Age	Pole position		d_p (°)	d_m (°)	References
			long. (°)	lat. (°)			
1	Lamotte Fm. (group I)	Late Cambrian	07 E	30 S	8	9	Al-Khafaji and Vincenz (1971)
2	Lamotte Fm. (group IIa)	Late Cambrian	12 E	38 N	4	8	Al-Khafaji and Vincenz (1971)
3	Lamotte Fm. (group IIb)	Late Cambrian	25 W	12 N	7	11	Al-Khafaji and Vincenz (1971)
4	Lamotte Fm. (locality E)	Late Cambrian	43 W	32 S	10	20	Al-Khafaji and Vincenz (1971)
5	Newfoundland	Late Cambrian	54 W	26 S	4	7	Beales et al. (1974)
6	Ratcliffe–Brook Fm.	Early Cambrian	56 W	10 S	14	21	Black (1964)
7	Wichita granites	Cambrian	33 W	2 S	6	9	Spall (1968)
8	Wichita Mountains	Cambrian	46 W	34 S	7	15	Ku et al. (1967)
9	Eradore Fm., Newfoundland	Early Cambrian	31 W	9 S	7	12	Black (1964)
10	Intrusives, Colorado	Cambrian-Ordovician	69 W	40 S	6	11	Larson and Mutschler (1971)
11	Wabana Group, Newfoundland	Early Ordovician	12 E	28 S	2	4	Deutsch and Rao (1970)
12	Trenton Limestone	Middle Ordovician	66 W	36 S	3	5	McElhinny and Opdyke (1973)
13	Beemerville Complex	Late Ordovician	54 W	35 S	9	18	Proko and Hargraves (1973)
14	St. George's Limestone	Early Ordovician	61 W	30 S	2	3	Deutsch and Rao (1976)
15	Quebec Group	Middle Ordovician	11 E	6 S	14	7	this study

reports of the area were published by Ells (1886, 1894). More recent geological work was done by Clark (1947, 1964a, b, 1970), Cooke (1952), Lespérance (1963) and Globensky (1969). This last worker subdivided the sedimentary sequence into five distinct units which are from the base to the top:

1 = Melbourne limestone (Lower Cambrian)
2 = sandstones and slates of the Granby group (Cambrian)
3 = Stanbridge slates (Upper Cambrian or Lower Ordovician)
4 = schists, sandstones and limestones of Normanskill age (Middle Ordovician)
5 = slates, siltstones and dolomites of the Trenton, Utica and Lorraine groups (Ordovician)

Riva (1966) assigned a Middle Ordovician age to unit *4* on the basis of palaeontological evidence.

Volcanic agglomerates and amygdaloidal volcanic rocks are interbedded in the black slaty schists of unit *4* with few interbeds of grey sandstone and dark-grey crystalline limestone. The volcanic agglomerate contains subrounded fragments with diameters of a few millimetres to many centimetres. The amygdaloidal and vesicular lavas, rarely pillowed and slightly recrystallized, are black to greenish grey on fresh surface and brownish on altered surface. Most of these lavas are andesites.

In the Actonvale area, Lespérance (1963) includes these volcanics in the Quebec Group. The lavas agglomerates and tuffs lie on top or are interbedded with grey massive limestones of Early to Middle Ordovician age. In their turn, these are underlain by grey and green slates, siltstones and calcareous sandstones. This last portion lies on top of the Sillery formation of red and green slates, green, red and grey arkoses, some conglomerates and slates. The volcanics are composed of plagioclase (An_{30}) almost

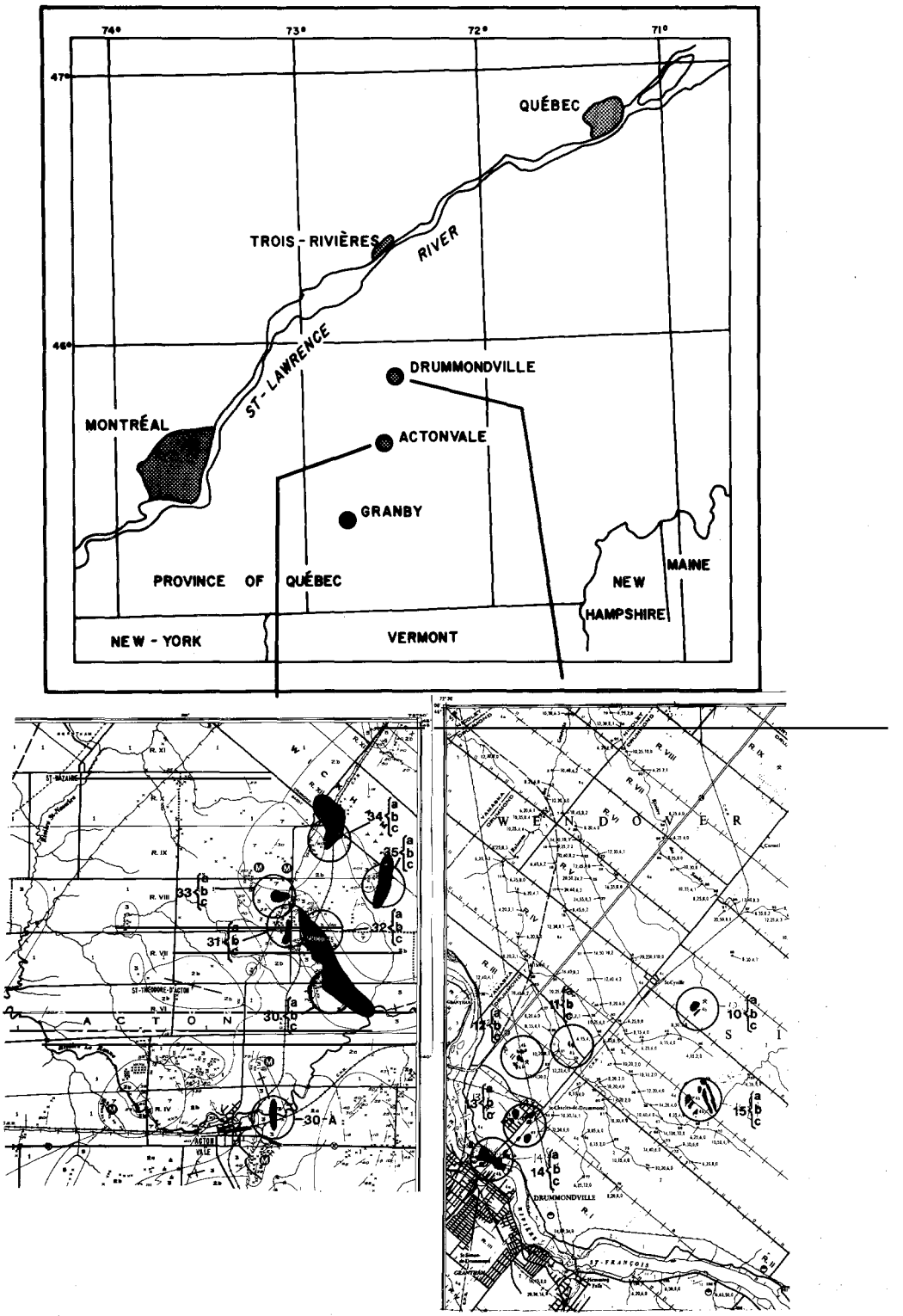


Fig. 1. Index map of the main localities and geological maps of the Actonvale and Drummondville areas with location of the sampling sites.

completely kaolinized; chlorite, epidote and some remanents of augite and magnetite are present. Some reddish lavas contain a large amount of hematite as an alteration product of magnetite; these lavas are highly magnetic. The source of these volcanic rocks has not been found but it is obviously located to the southeast, they are most likely related to the acidic tuffs of the Magog Group described by Ami (1900), McKay (1921) and St.-Julien (1970).

Several small intrusive bodies of diorite and gabbro are found in the Actonvale area. Seven are concentrated in a northeasterly trending zone between St.-Valérien and the NE corner of the area, and one near Roxton Falls in the SE corner. Similar intrusions of diorite and altered gabbro outcrop some 2,000 m to the east of the shoreline of the St.-Lawrence River and 3 km WSW of St.-Nicolas located to the SW of Quebec City. Whole-rock K–Ar dating by Shell Oil Co. yielded 450 ± 30 m.y. and 376 ± 25 m.y. respectively, indicating a Middle Ordovician age (Y. Gobensky, pers. commun., 1976); as far as the author is aware, no Ar loss was suspected.

2.2. Structural geology

In the Drummondville area, rocks of Normanskill age (unit *4*) and of the Saint-Germain Complex (unit *5*) form a fold plunging toward the southwest at an angle of about 6°. In the Actonvale area, the Champlain thrust trace (Logan's line) divides the Quebec Group to the southeast of the line from the Saint-Germain Complex to the northwest of the line. The rocks of the Saint-Germain Complex belong to the St.-Lawrence Lowlands Ordovician sequence. Way-up determinations in the Sillery green sandstones of the Quebec Group (Cambrian age) underlying the lavas indicate the presence of a major anticline trending northeasterly, SW of Actonvale. The folds are slightly overturned to the northwest, as the bedding generally dips steeply east.

3. Sampling procedure and measurements

36 samples (see Fig. 1) oriented with a Brunton or a solar compass were collected at eleven sites in the localities of Drummondville (5), Actonvale (4) and Grandy-Farnham (2) and drilled in the field with a portable diamond drill. The number of specimens cut in the laboratory varied between two and five per sample, yielding a total of 111 specimens. All but two outcrops of volcanic rocks mapped in the whole region were sampled.

The direction and intensity of remanent magnetization were measured with a Princeton Applied Research model SM-1 spinner magnetometer (sensitivity = 10^{-7} CGS e.m.u.) and for some very low intensity samples with a digital spinner magnetometer (model DSM-1) manufactured by Schonstedt Inst. Co. (sensitivity = 10^{-8} CGS e.m.u.). AF demagnetization was carried out to remove secondary components using a demagnetizer built at the University of Laval (maximum peak-to-peak intensity of 1,800 Oe) the performance of which was greatly improved by adding three large concentric mu-metal cylinders around the solenoid. Thermal treatment of the samples was done in a Schonstedt field-free thermal specimen demagnetizer (model TSD-1).

4. Thermal demagnetization

About 60% of the specimens were thermally demagnetized at 100, 200, 300, 400, 500, 550 and sometimes 600°C and approximately 70% of these specimens were relatively stable to thermal treatment. Magnetic instability was determined by the amount of variation of the intensity and orientation of the residual remanent vector. A plateau on the intensity coinciding with a minimum angular variation of the remanent vector was used to identify the stable residual component. The blocking temperatures of the magnetite-rich specimens are in the range 400–500°C, and those of hematite-rich specimens occur in the range of 550–600°C. After thermal demagnetization, the mean direction of the remanent orientations is $D = 92°, I = 15°$ after tilt correction. The palaeopole positions of 24 thermally demagnetized and stable samples is reversed and located at 191°E, 6°N ($d_m = 14°, d_p = 7°$). Table II lists the most important results.

5. AF demagnetization

A histogram of the NRM intensities (J_r) is presented in Fig. 2; the distribution is bimodal; the high-

TABLE II

Palaeomagnetic results for the lavas after thermal treatment

Sample No.	N	After fold test			Palaeopole position		Temperature (°C)
		D_2 (°)	I_2 (°)	α_{95} (°)	long. (°E)	lat. (°N)	
12A *1	3	138	66	17	344	39	400
12B *1	3	16	32	21	134	−23	400
12C *1	3	68	59	13	358	40	300
14A *1	3	115	20	9	352	−9	400
14B *1	3	69	−25	33	43	5	300
14C *1	2	81	4	1	24	7	300
30A *1	3	209	−28	32	239	−51	500
30B *1	3	128	−40	19	7	−43	300
30C *1	3	131	−40	15	5	−44	400
31A *1	3	57	19	12	39	32	400
31B *1	3	136	56	16	322	1	550
31C *1	3	97	56	42	346	21	500
32A *1	3	49	53	9	20	54	300
32B *1	4	145	55	35	314	−3	500
32C *1	4	5	−58	37	103	6	500
33A *1	3	38	−57	32	80	1	550
33B *2	3	31	9	15	66	42	500
33C *2	5	46	38	44	34	45	500
33D *2	3	127	59	16	325	8	300
34A *2	3	136	−74	17	62	−61	500
34B *1	3	87	23	42	13	13	550
34C *2	3	148	33	31	319	−20	500
34D *1	3	92	13	19	21	−5	300
34E *1	3	148	42	43	318	−15	500
Total and mean:							
24	75	92	15	19	11	−6	
					(d_m = 14°, d_p = 7°)		
					i.e., 191°E, 6°N (normal polarity)		

*1 Reverse polarities.
*2 Mixed polarities.

intensity values are found in hematite-bearing samples and the low-intensity values in magnetite-bearing samples. One or two pilot specimens from each sample were demagnetized between 50 and 700 Oe in steps of 50 or 100 Oe. Guided by the stepwise changes in remanence and intensity of the pilot specimen(s) of the samples, an objective "end-point" stability method was used in which the remaining specimens from the sample were AF demagnetized and their residual remanence measured (Symons and Stupavski, 1973). Pilot tests show that the AF intensity at which stable residual component is most thoroughly isolated is about 200–450 Oe (Fig. 3). Normalized curves for some typical test specimens from sites are shown in Fig. 4. Median destructive fields (MDF) are usually between 100 and 300 Oe. The magnetization directions during demagnetization of pilot specimens from three sites are presented in Fig. 5. All the other pilot specimens were AF demagnetized at successive intensities of 50–500 Oe in steps of 50 Oe, and 500–700 or 900 Oe in steps of 100 Oe. All the specimens indicated reverse or occasional mixed polarities – except two (11B and 13C) – which appear to have a normal polarity.

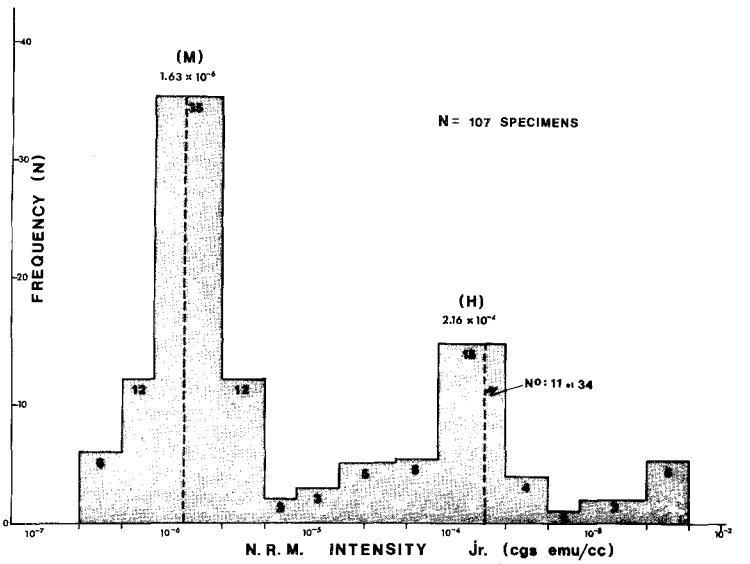


Fig. 2. Histogram of the NRM intensities of all the specimens of lavas.

6. Results of AF cleaned measurements

The NRM directions of the different samples of lavas investigated are moderately scattered. After AF treatment in the 200–450 Oe range depending on the nature of the samples, the dispersion within and between site generally decreased and the site mean direction became grouped about a ESE axis with equatorial inclination. The mean direction of the AF demagnetized NRM orientations is $D = 118°$, $I = 29°$ ($\alpha_{95} = 13°$; $k = 17$) with a corresponding pole position of 346°E, 7°S prior to tilt correction of the formations and $D = 129°$, $I = 7°$ ($\alpha_{95} = 12°$; $k = 25$) with a corresponding pole position of 344°E, 19°S

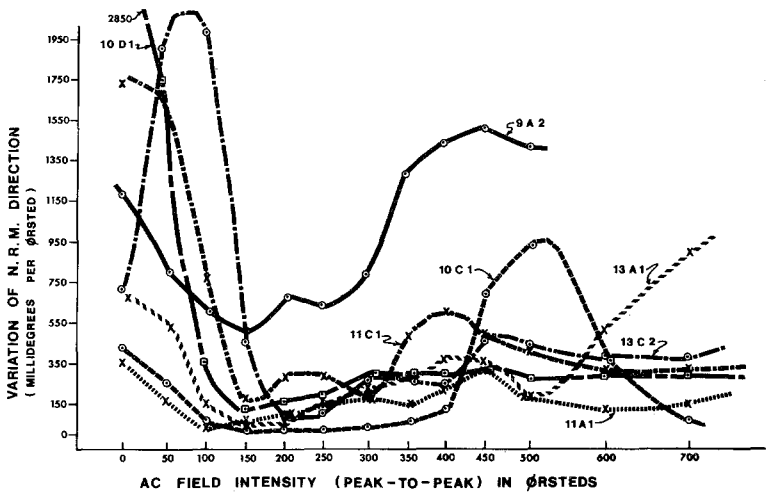


Fig. 3. Plot of the stability index for AF demagnetization of seven typical specimens of lavas.

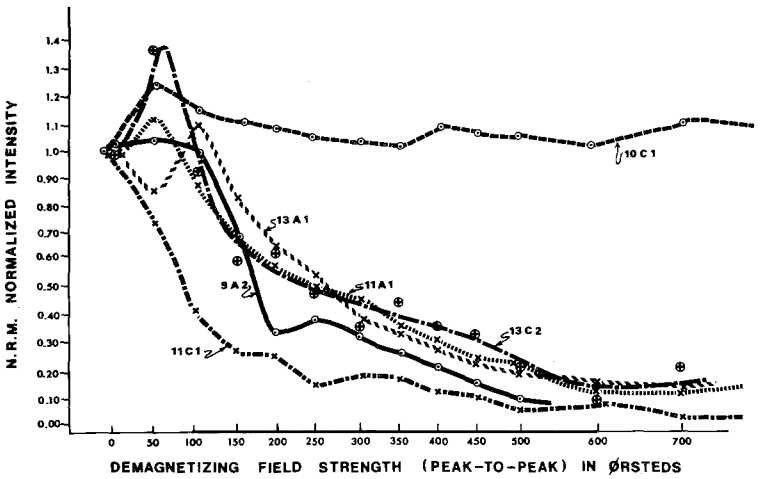


Fig. 4. Plot of normalized remanent intensity of six typical specimens as a function of the demagnetizing field strength.

after tilt correction of the formations. The dispersion of the poles is only slightly reduced after AF treatment and k (Fisher, 1953) does not increase substantially. Consequently, the fold test is not conclusive although the ancient remanence of the lavas investigated is demonstrated by departure of the site remanence directions from the present earth's field direction (342°, 70°N) and the evidence of polarity inversion at two sites. Fig. 6 shows the thermally and AF cleaned NRM directions which are believed to be stable on the above mentioned grounds.

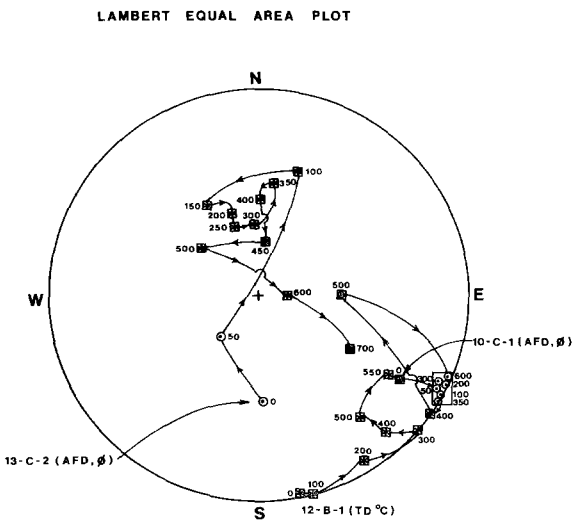


Fig. 5. Variation of the magnetization directions of two plot specimens of lavas as a function of the demagnetizing AF strength and heating temperature (after unfolding). ⊙ = upper hemisphere; * = lower hemisphere; □ = reverse polarity; A.F.D. = AF demagnetisation in Oersteds; TD = thermal demagnetisation in degrees Celsius.

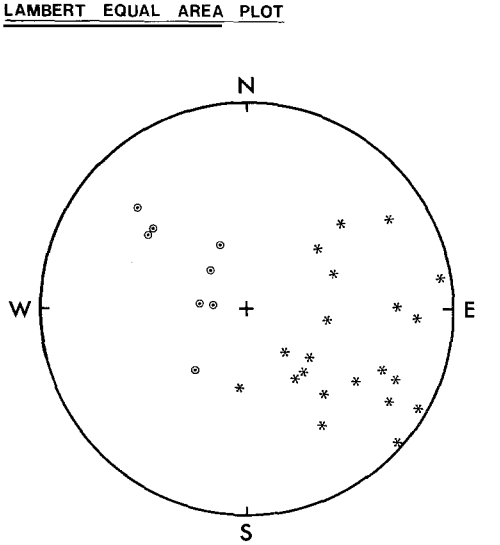


Fig. 6. Stereographic plot of NRM orientations of some 29 samples classified (tilt corrected as stable after AF cleaning and thermal treatment.

7. Age and origin of the magnetization

The palaeopole positions of the thermally and AF demagnetized lava samples are substantially different: 191°E, 6°N in the first case and 164°E, 19°N in the second. The pole position derived from thermally demagnetized samples lies about 30° from that derived from AF demagnetized samples. On the one hand, there is no reason, from the data available, to choose one pole over the other and on the other hand, it is quite wrong to combine AF and thermally-cleaned data because the thermal treatment may have removed a PTRM unaffected by the AF treatment. The palaeomagnetic poles from AF and thermally-cleaned data should be treated separately and analyzed accordingly.

We will first make an attempt to assess the data before and after tilt corrections. The thermal-cleaned result, which yields a pole position of 191°E, 6°N (11°E, 6°S) after tilt correction, is believed to be primary; this palaeopole is the nearest to Lower Palaeozoic poles, in particular the one of the Lamotte Formation (group *1*; pole position: 7°E, 29°S) which is of Late Cambrian age and the one of the Wabana Group, Newfoundland (pole position: 12°E, 28°S) which is of Early Ordovician age (McElhinny, 1973). The palaeomagnetic direction of the thermal-cleaned data is definitely different from the post-Ordovician apparent wander path for North America prior to tilt correction, implying that this direction is most probably primary. In any event, this palaeomagnetic direction differs from poles appropriate to the interval of folding in Siluro-Devonian times.

The AF cleaned direction (prior to tilt correction; 165°E, 7°N) coincides somewhat with poles expected from the latest Precambrian–Early Cambrian age but not at all with poles expected from the time of folding (i.e. Mid-Ordovician Taconic orogen). This conclusion leads to some kind of a dilemma. On the one hand, one can suppose that the AF cleaned direction (after correction; 164°E, 19°N) could be primary but then the age of the volcanics would be latest Precambrian–Early Cambrian. This implies that the volcanic units (blocks) of this study were incorporated (upthrusted, imbricated or wedged) in the Ordovician sediments during the Taconian orogen. They could then be fragments of Lower Palaeozoic flood basalts such as the Lighthouse Cove Formation and Belle Isle in Newfoundland and eventually the Tibbit Hill basal volcanic member of the Oak Hill sequence in the Eastern Townships of Quebec. On the other hand, one can argue that the AF cleaning was ineffective and consequently only the thermal-cleaned results are valuable. Petrological and scarce geochemical data indicate that the metalavas of this study and the ones of the plateau basalts are not comparable.

For this reason, the second alternative appears more realistic and probable; the tilt corrected pole of the thermal-cleaned results which appears to be primary suggests that the apparent polar-wandering path for North America is considerably more complicated than suggested by present data (Fig. 7).

The average of the four poles from Cambrian rocks in Norht America reported in the literature (Table I) is 34°W, 10°S while the average of six poles from Ordovician rocks (Table I) is 49°W, 32°S. The thermally cleaned and tilt-corrected palaeopole (11°E, 6°S) obtained in this study does not lie close to any of the poles determined from Cambrian and Ordovician rocks known to be on the stable part of the North American craton but is nearer to the one of the Wabana Group, Newfoundland. However, this last Lower Ordovician pole is from eastern Newfoundland and may not have been part of the North American plate. The palaeopole of this study is located some 15° to the west and 15° to the north of the New-

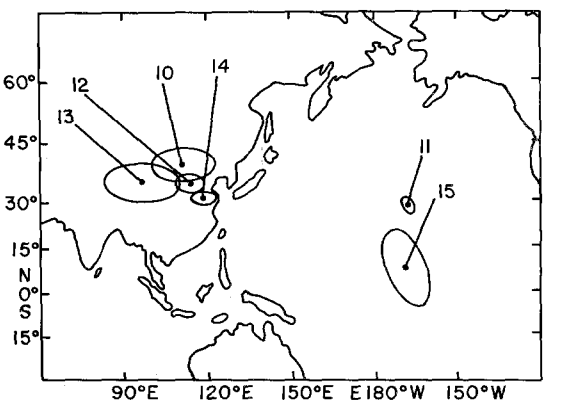


Fig. 7. Pole position with 95% confidence ovals (Table I) for rock suites of Ordovician age on the North American continent (mainland and Newfoundland). The number refer to the ones listed in Table I.

TABLE III

Palaeomagnetic results for the lavas after AF demagnetization

Sample No.	N	Before fold test			After fold test			Field intensity (Oe)
		D_1 (°)	I_1 (°)	α_{95} (°)	D_2 (°)	I_2 (°)	α_{95} (°)	
9A *	2	124	43	19	133	−27	18	200
9B *	3	82	50	37	132	2	25	200
10A *	4	119	34	2	122	17	2	400
10B *	3	107	42	2	114	26	1	400
10C *	3	119	18	5	120	1	4	400
10D *	3	117	51	8	122	34	9	400
11A *	3			unstable				
11B	3	302	35	27	252	25	24	350
11C *	2			unstable				
13A *	3	195	−52	17	275	−79	16	400
13B *	4	113	−61	26	46	−55	24	400
13C	3	210	−75	3	339	−64	4	250
Total and mean:								
12	36	118	29	13	129	7	12	
Pole position:		346°, 7° S			344°, 19° S			
		(d_m = 11.5°, d_p = 6.1°, k = 17) i.e. 164° E, 19° N						

* Reverse polarities.

TABLE IV

Cambrian and Ordovician poles from the Caledonides

Entry *1	Age *2	Palaeopole		α_{95} (°)	d_m (°)	d_p (°)
		long. (° E)	lat. (° N)			
2	C_1	169	29		6	10
15	$C_u - O_1$	129	15		15	27
4	O_1	168	11	10		
6	O_1	189	11		6	10
8	O_1	176	9		4	7
11	O_1	179	16		6	11
13	O_1	172	5		19	27
14	O_1	119	2		16	28
19	O_1	218	11		10	18
	Mean	174	9		10	18
12	O_u	182	2° S		11	16
20	O_u	221	42	–		
21	O_u	117	37	–		
22	O_u	164	33	–		
9	O_u(?)	188	23		8	15
	Mean	174	27		10(?)	15(?)
23	$O_u - S_1$	154	32		9	17

*1 Entry number after Briden et al. (1973).

*2 C_1 = Lower Cambrian; C_u = Upper Cambrian; O_1 = Lower Ordovician; O_u = Upper Ordovician; S_1 = Lower Silurian.

foundland result. Consequently, a post-Ordovician rotation of Newfoundland with respect to the mainland by an angle of 25° is very unlikely even though possible at the extreme limit.

8. Comparison with palaeomagnetic data from Europe

The available palaeopole positions for Cambro-Ordovician times of the Caledonides (Briden et al., 1973) are listed in chronological sequence in Table IV. The mean Lower Ordovician palaeopole (174°E, 9°N) for Europe is not significantly different from the one obtained in this study (191°E, 6°N) and two of these seven poles, i.e. entries 6 and 19 of Table IV (positions: 189°E, 11°N and 218°E, 11°N respectively) are particularly close. This implies that at least at Early Ordovician times, the mean poles from Europe and this pole from North America occupied a roughly similar position and on the basis of the present evidence; the broad agreement between the Lower Palaeozoic poles from Europe and North America may not be regarded as fortuitous.

9. Discussion of the results

It was originally intended to select AF cleaned pole on the basis of its apparent greater stability until it was noticed that this apparent stability could imply that the AF demagnetization was less effective in removing secondary magnetizations. For this reason, the thermal-cleaned result was considered as closer to the true value of the primary magnetization. The reason why this pole does not fall on the tentatively but poorly-established apparent polar-wander path of Cambrian and Ordovician rocks of the North American plate is obscure. It could be that some of the samples included in the final mean were not perfectly cleaned or that they possess mainly present-day magnetic field direction as was observed in a few (six) AF cleaned samples which were rejected but this does not appear to be the case for the other AF cleaned samples and for the thermally treated samples. Alternatively, it is not impossible that the Ordovician rocks of the Appalachian sector (allochtonous zone) of southern Quebec did not constitute an integral part of the North American plate (autochtonous zone in the Quebec region) but instead formed part of a rotated block possibly related to the eastern Newfoundland block. In this respect, it is worthwhile to point out a similar situation whereby a substantial pole divergence is noted between the Upper Cambrian–Lower Ordovician pole position given by Beales et al. (1974) from western Newfoundland and the uppermost Lower Ordovician pole position presented by Deutsch and Rao (1970) from eastern Newfoundland.

In spite of the fact that the lavas investigated in this study are relatively altered, the palaeopole is more likely to be related to the time of extrusion of the lavas which is almost contemporaneous to the Taconic orogeny (Middle Ordovician time) than to the imprinting of a later tectonic event on these lavas (e.g., Acadian orogeny in Siluro-Devonian time). The poles of this study differ definitely from the Siluro-Devonian pole before tilt correction of the palaeomagnetic data. Indeed, Silurian and Devonian poles have longitudes and latitudes which are located respectively further west and further north than the Ordovician poles. Another interesting fact is that most known Silurian and Lower Devonian pole positions display a reversed polarity which is not the case in this study. Consequently, in spite of relatively large tectonic tilts, scatter in the directions of residual remanence and conceivable remagnetization in some of the samples, these palaeomagnetic data tell a significant story.

Acknowledgments

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